
ME 622: Biomechanics of Musculoskeletal System

Course Code	ME 622
Course Name	Biomechanics of Musculoskeletal System
L-T-P-C	3-0-0-3
Prerequisites	ME206 Mechanics of solids, IC240 Mechanics of Rigid Bodies , IC242 Continuum Mechanics , ME352 Finite Element Methods i n Engineering
Intended for	UG/PG
Distribution	Elective
One-time approval	9th Senate (May 2016)

Course Contents

- **Module I**
Introduction to Biomechanics, Basic terminology and concept of hu man musculoskeletal system, anatomy and overall function. [4 Lectures]
- **Module II**
Biomechanics of Tissues and Structures of musculoskeletal system composition and structure. [3 Lectures]
- **Module III**
Biomechanical behaviour bone, articular cartilage, muscle, tendon and ligament. [8 Lectures]
- **Module IV**
Biom echanics of joints structure, range of motions, musculoskeletal model of forces: (i) hip; (ii)knee; (iii) shoulder; (iv) elbow; (v) spine, lubrication of joints. [8 Lectures]
- **Module V**
Motion and gait analysis method, gait cycle, segmental kinetics, engineering apprao ches to posture analysis. [6 Lectures]
- **Module VI**
Joint replacement and fracture fixation stress analysis and basic design approach, failure mechanisms, wear in joint arthroplasty, bone remodelling. [8 Lectures]
- **Module VII**
Biomaterials properties and application. [2 Lectures]
- **Module VIII**
Image Processing T echniques: Quantitative CT scan / MRI reconstruction and generation of solid (CAD) models. [3 Lectures]

Text Books

1. M. Nordin and V.H. Frankel, **Basic Biomechanics of the Musculoskeletal System**, 3rd edition, LWW.
2. Y. C. Fung, **Biomechanics**, 2nd Edition, Springer
3. J.D. Humphery and S.L. Delange, **An Introduction to Biomechanics**, Springer

References

1. Dominique G. Poitout, **Biomechanics and Biomaterials in Orthopedics**, by Publisher: Springer
2. Kelvin L. Ong, Scott Lovald, Jonathan Black, **Orthopaedic Biomaterials in Research and Practice**, 2nd Edition, CRC Press.